2 LOGIC_GND Logická zem
3 NC Rezerva
4 NC Rezerva
5 24V_BATT Batériová vetva 24 V
6 5V_BATT Batériová vetva 5 V
7 BATT_GND Batériová zem
8 BATT_GND Batériová zem
9 5V_LOGIC LOGIC 5 V
0 LOGIC_GND Logická zem
1 NC Rezerva
2 NC Rezerva
3 I2C_SCL I²C hodinový signál
4 I2C_SDA I²C dátový signál
5 NC Rezerva pre budúce IRQ
6 NC Rezerva
7 SPI_SCK SPI clock
8 LOGIC_GND Referenčná zem SPI
9 SPI_MOSI SPI controller → periféria
0 SPI_MISO SPI periféria → controller
1 SPI_CS0 Výber SPI zariadenia 0
2 SPI_CS1 Výber SPI zariadenia 1
3 SPI_DC Display Data/Command
4 LOGIC_GND Referenčná zem SPI
5 NC Rezerva
6 RS485_GND Referenčná zem RS-485
7 RS485_A Diferenciálna linka A
8 RS485_B Diferenciálna linka B
9 NC Rezerva
0 NC Rezerva

BACKBONE V2.1
Medulla Oblongata

H101
M20-6102045

24V 1 2 24V
GND 3 4 GND
5V 5 6 5V
7 8
9 10
LOGIC_GND 11 12 LOGIC_GND
13 14
24V_BATT 15 16 5V_BATT
BATT_GND 17 18 BATT_GND
5V_LOGIC 19 20 LOGIC_GND
21 22
I2C_SCL 23 24 I2C_SDA
25 26
SPI_SCK 27 28 LOGIC_GND
SPI_MOSI 29 30 SPI_MISO
SPI_DC 31 32 SPI_CS2
SPI_SC1 33 34 LOGIC_GND
UART_TX 35 36 RS485_GND
RS485_B 37 38 RS485_A
UART_RX 39 40 LOGIC_GND

R120
MBB02070Z0000ZC100
24V_LOGIC

MANUS

IN LINE PULL UP ULNA

R102 ERJPA3F4701V
LOGIC_3V3
R103 ERJ-3EKF1001V
INDEX
C102 C0603C103K5RACTU
LOGIC_GND

R104 ERJPA3F4701V
LOGIC_3V3
R105 ERJ-3EKF1001V
POLLEX
C103 C0603C103K5RACTU
LOGIC_GND

R106 ERJPA3F4701V
LOGIC_3V3
R107 ERJ-3EKF1001V
DIGITUS_MEDIUS
C104 C0603C103K5RACTU
LOGIC_GND

R108 ERJPA3F4701V
LOGIC_3V3
R109 ERJ-3EKF1001V
DIGITUS_ANULARIS
C105 C0603C103K5RACTU
LOGIC_GND

DIGITUS MINIMUS

CN101
1990067
LOGIC_GND ×8

0-30V OUT LINE RADIUS

Pull-down na GPIO
R112 ERJ2RKF4702X
LOGIC_GND
Obmedzenie prúdu LED
R110 ERJ3EKF3300V
OS_Trapenzium
U101 AQY211EH
LOGIC_GND
D101 SMBJ26CA_C83332

Pull-down na GPIO
R111 ERJ2RKF4702X
LOGIC_GND
Obmedzenie prúdu LED
R113 ERJ3EKF3300V
OS_Trapezoideum
U102 AQY211EH
LOGIC_GND
D102 SMBJ26CA_C83332

Pull-down na GPIO
R114 ERJ2RKF4702X
LOGIC_GND
Obmedzenie prúdu LED
R115 ERJ3EKF3300V
OS_Capitatum
U104 AQY211EH
LOGIC_GND
D103 SMBJ26CA_C83332

Pull-down na GPIO
R116 ERJ2RKF4702X
LOGIC_GND
Obmedzenie prúdu LED
R117 ERJ3EKF3300V
OS_Hamatum
U105 AQY211EH
LOGIC_GND
D104 SMBJ26CA_C83332

CN102
1990067
1 2 3 4 5 6 7 8

ESP32-S3-WROOM-1U-N16R8

ADC1
ADC2
DI1_LOGIC POLLEX
DI2_LOGIC INDEX
DI3_LOGIC DIGITUS_MEDIUS
DI4_LOGIC DIGITUS_ANULARIS
DI_AUX1
DO1_CMD OS_Trapenzium
DO2_CMD OS_Trapezoideum
DO3_CMD OS_Capitatum
DO4_CMD OS_Hamatum
DO_FAULT_INT
ESP_BOOT
RS485_A
RS485_B
LOGIC_GND
LOGIC_3V3

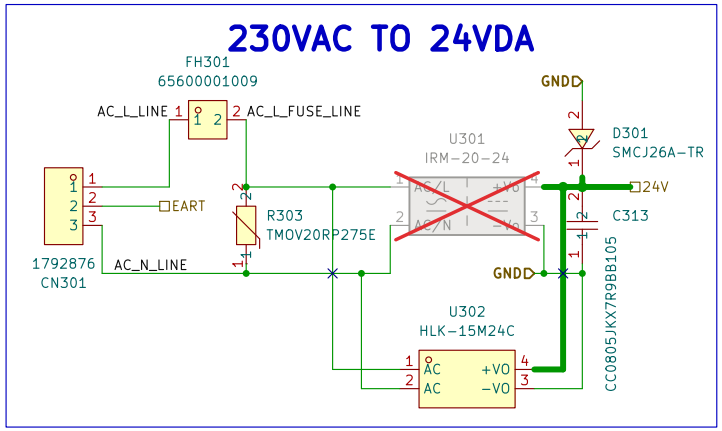
I2C_SCL
I2C_SDA
JTAG TDI
JTAG TMS
RGB_DATA
SPI_CS1 SPI_SC1
SPI_CS2 SPI_CS2
SPI_MISO SPI_MISO
SPI_MOSI SPI_MOSI
SPI_SCK SPI_SCK
SPI_DC SPI_DC
UART0_RX
UART0_TX
UART_RX
UART_TX
CHASIS CHASSIS_GND

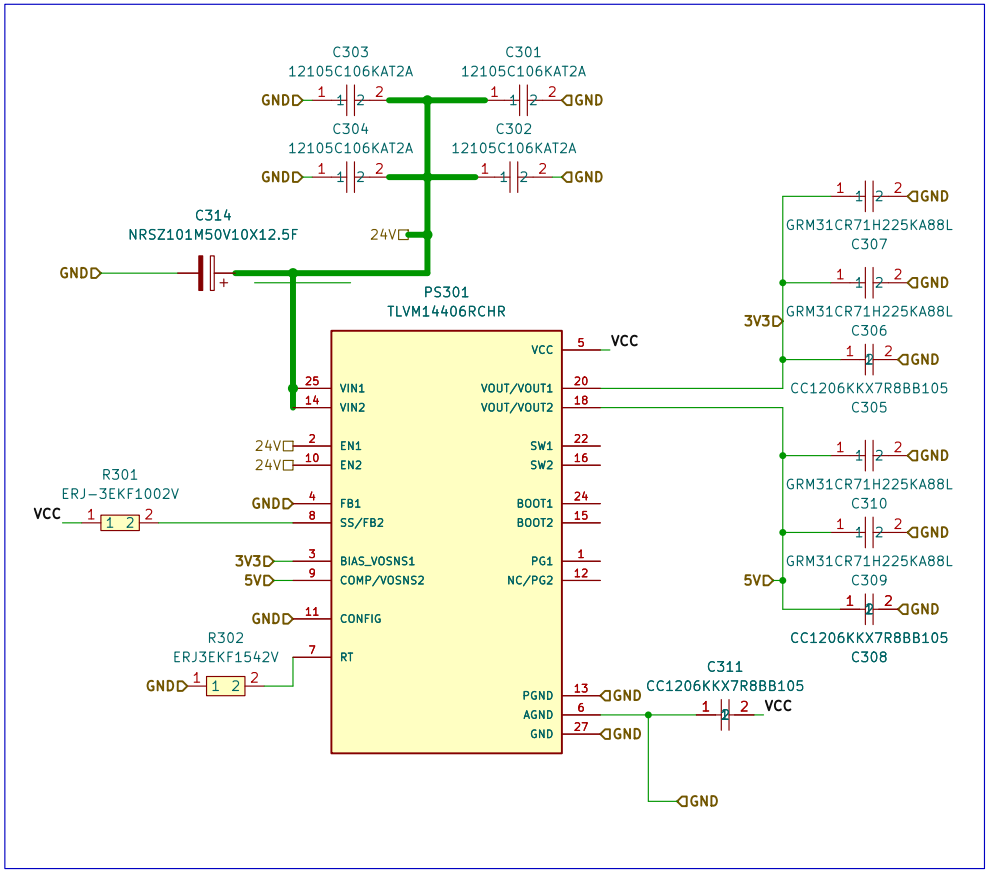
R121 MBB02070Z0000ZC100 36V_TO_5V+3V3
LOGIC_3V3 3V3
R119 MBB02070Z0000ZC100
5V_LOGIC 5V
24V_LOGIC 24V
CHASSIS_GND
C107 DE1E3RA102MA4BN01F
R122 CRCW12061M00FKEA
LOGIC_GND
EART
GND

RGB

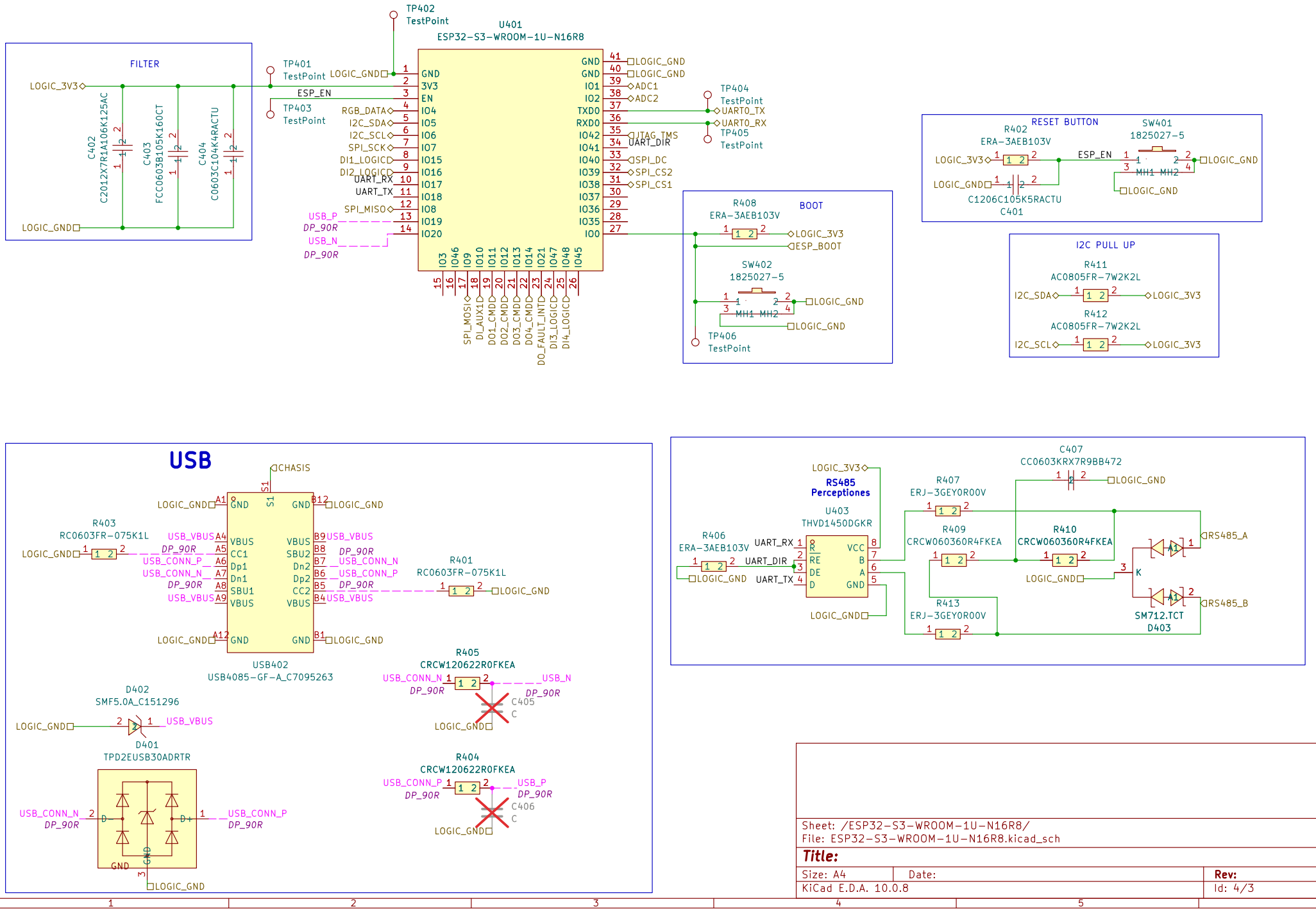
LOGIC_GND
U103 SN74AHCT1G125DBVR
C101 C0805C104K5RACTU_C3018562
5V_LOGIC
RGB_DATA
R118 ERA-3AEB103V
R101 ERJ3EKF3300V
CN104 1989751

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Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 1/3





Sheet: /36V_TO_5V+3V3/		
File: 36v to 5v + 3v3.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 3/3


FILTER
LOGIC_3V3
LOGIC_GND
C402
C2012X7R1A106K125AC
C403
FCC0603B105K160CT
C404
C0603C104K4RACTU
TP402
TestPoint
U401
ESP32-S3-WROOM-1U-N16R8
TP401
TestPoint
TP403
TestPoint
ESP_EN
RGB_DATA
I2C_SDA
I2C_SCL
SPI_SCK
DI1_LOGICD
DI2_LOGICD
UART_RX
UART_TX
SPI_MISO
USB_P
DP_90R
USB_N
DP_90R
TP404
TestPoint
UART0_TX
UART0_RX
TP405
TestPoint
JTAG_TMS
UART_DIR
SPI_DC
SPI_CS2
SPI_CS1
ADC1
ADC2
SPI_MOSI
DI_AUX1D
DO1_CMDD
DO2_CMDD
DO3_CMDD
DO4_CMDD
DO_FAULT_INTD
DI3_LOGICD
DI4_LOGICD
R408
ERA-3AEB103V
BOOT
LOGIC_3V3
ESP_BOOT
SW402
1825027-5
MH1
MH2
TP406
TestPoint
RESET BUTTON
R402
ERA-3AEB103V
SW401
1825027-5
C1206C105K5RACTU
C401
I2C PULL UP
R411
AC0805FR-7W2K2L
R412
AC0805FR-7W2K2L
USB
CHASIS
USB402
USB4085-GF-A_C7095263
R403
RC0603FR-075K1L
R401
RC0603FR-075K1L
R405
CRCW120622R0FKEA
C405
R404
CRCW120622R0FKEA
C406
D402
SMF5.0A_C151296
D401
TPD2EUSB30ADRTR
RS485
Perceptiones
U403
THVD1450DGKR
R406
ERA-3AEB103V
R407
ERJ-3GEY0R00V
R409
CRCW060360R4FKEA
R410
CRCW060360R4FKEA
R413
ERJ-3GEY0R00V
C407
CC0603KRX7R9BB472
SM712.TCT
D403
RS485_A
RS485_B
Sheet: /ESP32-S3-WROOM-1U-N16R8/
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Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.8
Id: 4/3